

Nonlocal DE

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1 MTEGR Framework

$$H^2 = \frac{8\pi G}{3(1+S)}(\rho_M + \rho_{DE}) \quad (1)$$

$$\frac{d\rho}{dt} = -3(\rho + P)H - \rho \frac{d \log(1+S)}{dt} \quad (2)$$

Now if $\frac{P}{\rho} = \omega$ that is constant in time we solve Equation 2 and we have

$$\rho = \rho_o \frac{(1+S_0)^2}{(1+S)} (1+z)^{3(1+\omega)} \quad (3)$$

With $\rho_o = \rho_o^{\Lambda CDM}$

$$2\frac{\ddot{a}}{a} + H^2 = -8\pi G \frac{P}{1+S} - 2\frac{d \log(1+S)}{dt} H \quad (4)$$

$$\bar{\rho}_{DE} = \frac{\rho_M}{1+S} - \bar{\rho}_M + \frac{\rho_{DE}}{1+S} \quad (5)$$

$$\bar{\rho}_{DE} = \rho_{M0} \left(\left(\frac{1+S_0}{1+S} \right)^2 - 1 \right) (1+z)^3 + \rho_{DE0} \left(\frac{1+S_0}{1+S} \right)^2 (1+z)^{3(1+w_{DE})} \quad (6)$$

$$\bar{P}_{DE} = w_{DE} \rho_{DE0} \left(\frac{1+S_0}{1+S} \right)^2 (1+z)^{3(1+w_{DE})} + \frac{1}{4\pi G} \frac{d \log(1+S)}{dt} H. \quad (7)$$

$$\bar{w}_{DE} = \frac{w_{DE} \left(\frac{1+S_0}{1+S} \right)^2 (1+z)^{3(1+w_{DE})} + \frac{1}{4\pi G} \frac{d \log(1+S)}{dt} \frac{H}{\rho_{DE0}}}{\left(\frac{1+S_0}{1+S} \right)^2 (1+z)^{3(1+w_{DE})} + \frac{\rho_{M0}}{\rho_{DE0}} \left(\left(\frac{1+S_0}{1+S} \right)^2 - 1 \right) (1+z)^3}. \quad (8)$$

if we solve

$$\begin{aligned} \frac{1}{4\pi G} \frac{d \log(1+S)}{dt} \frac{H}{\rho_{DE0}} &= \frac{1}{4\pi G} \frac{d \log(1+S)}{dt} \frac{H^2}{\rho_{DE0}} \frac{a}{\dot{a}} \\ &= \frac{1}{4\pi G} \frac{d \log(1+S)}{da} \frac{H^2}{\rho_{DE0}} a \\ &= \frac{2}{3} \frac{d \log(1+S)}{da} \frac{a}{(1+S)} \left(\frac{\rho_M}{\rho_{DE0}} + \frac{\rho_{DE}}{\rho_{DE0}} \right) \\ &= \frac{2a}{3} \frac{d \log(1+S)}{da} \left(\frac{\rho_{M0}}{\rho_{DE0}} \left(\frac{1+S_0}{1+S} \right)^2 (1+z)^{3(1+\omega_M)} + \left(\frac{1+S_0}{1+S} \right)^2 (1+z)^{3(1+\omega_{DE})} \right) \end{aligned} \quad (9)$$

we also know $\frac{d}{da} = -(1+z)^2 \frac{d}{dz}$ and also use $\frac{\rho_{M0}}{\rho_{DE0}} \simeq \frac{1}{2}$ then we have

$$\frac{1}{4\pi G} \frac{d \log(1+S)}{dt} \frac{H}{\rho_{DE0}} = -\frac{2}{3} (1+z) \frac{d \log(1+S)}{dz} \left(\frac{1}{2} \left(\frac{1+S_0}{1+S} \right)^2 (1+z)^{3(1+\omega_M)} + \left(\frac{1+S_0}{1+S} \right)^2 (1+z)^{3(1+\omega_{DE})} \right) \quad (10)$$

then if we use $\omega_{DE} = -1$ then for $\bar{w}$

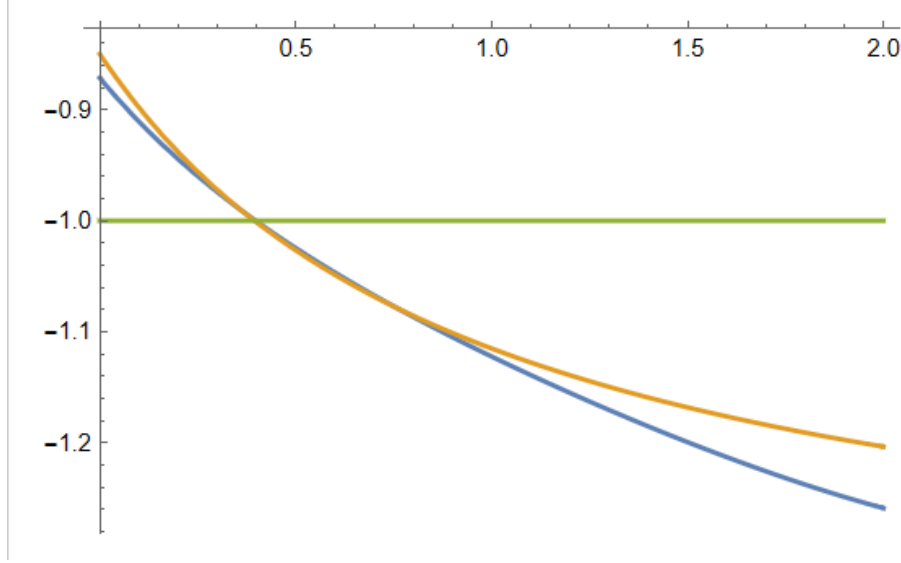


Figure 1: Green in Λ CDM Blue is nonlocal S_2 whit value $\alpha = 0.156573, \beta = -1.91 - 1.695$ and orange is CPL with DESI DR2 $\omega_0 = -0.850, \omega_1 = -0.53$

$$\bar{w}_{\text{DE}} = \frac{-\left(\frac{1+S_0}{1+S}\right)^2 - \frac{2}{3}(1+z)\frac{d\ln(1+S)}{dz}\left(\frac{1}{2}\left(\frac{1+S_0}{1+S}\right)^2(1+z)^3 + \left(\frac{1+S_0}{1+S}\right)^2\right)}{\left(\frac{1+S_0}{1+S}\right)^2 + \frac{1}{2}\left(\left(\frac{1+S_0}{1+S}\right)^2 - 1\right)(1+z)^3} \quad (11)$$

then factor out $\left(\frac{1+S_0}{1+S}\right)^2$

$$\bar{w}_{\text{DE}} = \frac{-1 - \frac{2}{3}(1+z)\left(\frac{1}{2}((1+z)^3 + 1)\frac{d\ln(1+S)}{dz}\right)}{1 + \frac{1}{2}\left(1 - \left(\frac{1+S_0}{1+S}\right)^2\right)(1+z)^3} \quad (12)$$

The alternative $s(z)$ can be generated with

$$s(z)_2 = 0.03\left((1+z)^{-2}\right) + \alpha\left((1+z)^\beta\right) \quad (13)$$

2 Dynamics of Fluids for MTEGR

$$H^2 = \frac{8\pi G}{3(1+s)} \sum \rho_i \quad (14)$$

$$2\frac{\ddot{a}}{a} + H^2 = \frac{-8\pi G}{1+S} \sum P_i - 2H \frac{d \ln(1+S)}{dt} \quad (15)$$

Let us compute the derivative from Equation 14:

$$2\frac{\dot{a}}{a} \left(\frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2} \right) = -\frac{8\pi G}{3(1+s)} \frac{d \log(1+s)}{dt} \sum \rho_i + \frac{8\pi G}{3(1+s)} \sum \frac{d\rho_i}{dt} \quad (16)$$

Then, we compute the derivative from Equation 14, using Equation 15 in Equation 16.

$$\begin{aligned} \frac{\dot{a}}{a} \left(\frac{-8\pi G}{1+S} \sum P_i - 2H \frac{d \ln(1+S)}{dt} - 3 \frac{\dot{a}^2}{a^2} \right) = \\ -\frac{8\pi G}{3(1+s)} \frac{d \log(1+s)}{dt} \sum \rho_i + \frac{8\pi G}{3(1+s)} \sum \frac{d\rho_i}{dt} \end{aligned} \quad (17)$$

Then, to clarify, we find that

$$\sum_i \left[\frac{d\rho_i}{dt} + 3\frac{\dot{a}}{a}(\rho_i + P_i) + \frac{d \log(1+s)}{dt} \rho_i \right] = 0 \quad (18)$$

We can take the derivative with respect to a , and then we obtain

$$\sum_i \left[\frac{d\rho_i}{da} + 3\frac{1}{a}(\rho_i + P_i) + \frac{d \log(1+s)}{da} \rho_i \right] = 0 \quad (19)$$

We also find that

$$\frac{d\rho_i}{dt} + 3\frac{\dot{a}}{a}(\rho_i + P_i) + \frac{d \log(1+s)}{dt} \rho_i = 0,$$

by assuming that these fluids are non-interacting, meaning that $\rho_i = \rho_i(a) \neq \rho_i(a, \rho_j, \dots)$. This implies that the solution for one fluid is independent of the existence of the second fluid. Therefore, we have

$$\frac{d\rho_i}{dt} + 3\frac{\dot{a}}{a}(\rho_i + P_i) + \frac{d \log(1+s)}{dt} \rho_i = 0 \quad (20)$$

and if we solve this

$$\rho_i(t) = \rho_{i0} \frac{1+S(t_0)}{1+S(t)} a(t)^{-3(1+w_i)} \quad (21)$$

3 Dynamical fluid for MTEGR

Equation (21) provides the solution for a fluid with a constant equation of state parameter, i.e., $\frac{P_i}{\rho_i} = \omega_i$, known as a barotropic fluid. However, if we aim to describe a more general dynamical fluid where the equation of state varies with redshift as $\frac{P}{\rho} = f(z)$, then the energy conservation equation must be modified accordingly.

Starting from the generalized continuity equation:

$$\frac{d\rho}{dt} + 3\frac{\dot{a}}{a}(1 + f(z))\rho + \frac{d \log(1 + s)}{dt}\rho = 0, \quad (22)$$

we perform a change of variables from cosmic time t to redshift z using the relation

$$\frac{dt}{dz} = \frac{-1}{\dot{a}(1 + z)^2}.$$

Multiplying both sides of equation (22) by $\frac{dt}{dz}$, we obtain:

$$\frac{d\rho}{dz} - 3\frac{1 + f(z)}{1 + z}\rho + \frac{d \log(1 + s)}{dz}\rho = 0. \quad (23)$$

This is a first-order linear differential equation in $\rho(z)$. Integrating both sides gives:

$$\begin{aligned} \int dz \left[\frac{d \log \rho}{dz} - 3\frac{1 + f(z)}{1 + z} + \frac{d \log(1 + s)}{dz} \right] &= 0 \\ \Rightarrow \log \rho - 3 \int_0^z \frac{1 + f(z')}{1 + z'} dz' + \log(1 + s(z)) &= \text{const.} \end{aligned} \quad (24)$$

Exponentiating both sides and absorbing the constant into a normalization factor $\hat{\rho}_0$, we obtain:

$$\rho(z) = \hat{\rho}_0 \frac{(1 + s_0)^2}{1 + s(z)} \exp \left[3 \int_0^z \frac{1 + f(z')}{1 + z'} dz' \right], \quad (25)$$

where the constant $\hat{\rho}_0$ is fixed by requiring consistency with the standard Λ CDM model at $z = 0$, as in equation (40).

To proceed further, we adopt the CPL parametrization:

$$f(z) = \omega_0 + \omega_a \frac{z}{1 + z}.$$

We now evaluate the integral appearing in equation (25) step by step:

$$\begin{aligned} \int_0^z \frac{1 + f(z')}{1 + z'} dz' &= \int_0^z \frac{1 + \omega_0 + \omega_a \frac{z'}{1 + z'}}{1 + z'} dz' \\ &= (1 + \omega_0) \int_0^z \frac{1}{1 + z'} dz' + \omega_a \int_0^z \frac{z'}{(1 + z')^2} dz' \\ &= (1 + \omega_0) \ln(1 + z) + \omega_a \left[\ln(1 + z) + \frac{1}{1 + z} - 1 \right] \\ &= (1 + \omega_0 + \omega_a) \ln(1 + z) - \omega_a \left(\frac{z}{1 + z} \right). \end{aligned}$$

Substituting this result into equation (25), we find the final expression for the density:

$$\rho(z) = \hat{\rho}_0 \frac{(1 + s_0)^2}{1 + s(z)} (1 + z)^{3(1+\omega_0+\omega_a)} \exp \left[-3\omega_a \left(\frac{z}{1+z} \right) \right]. \quad (26)$$

We observe that setting $\omega_0 = -1$, $\omega_a = 0$, and $s(z) = 0$ recovers the Λ CDM.

Finally, we note that any proposed function $f(z)$ must also be consistent with the second Friedmann equation (equation (4)) to ensure a fully self-consistent cosmological model.

4 MCMC for MTEGR

4.1 Markov Chain Monte Carlo and the Metropolis Algorithm

Markov Chain Monte Carlo (MCMC) methods are a class of algorithms used to sample from probability distributions when direct sampling is infeasible. One of the foundational MCMC methods is the **Metropolis algorithm**, introduced by Metropolis et al. in 1953. This algorithm constructs a Markov chain whose stationary distribution is the target posterior distribution $P(\theta | D)$, where θ denotes the parameter vector and D the observed data.

4.1.1 The Metropolis Algorithm

The Metropolis algorithm generates a sequence of samples $\{\theta^{(i)}\}$ by the following steps:

1. **Initialize:** Choose an initial parameter vector $\theta^{(0)}$.
2. **Propose a new state:** At iteration i , propose a new parameter vector θ' from a proposal distribution $q(\theta' | \theta^{(i)})$. A common choice is a symmetric Gaussian centered at the current state:

$$\theta' = \theta^{(i)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \Sigma) \quad (27)$$

$$q(\theta' | \theta^{(i)}) = \frac{1}{\sqrt{(2\pi)^d \det(\Sigma)}} \exp\left(-\frac{1}{2}(\theta' - \theta^{(i)})^\top \Sigma^{-1}(\theta' - \theta^{(i)})\right) \quad (28)$$

- Σ : Covariance matrix of the proposal distribution. It is often chosen to be diagonal:

$$\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_d^2)$$

where σ_i^2 represents the variance along the i -th parameter direction.

- d : Dimensionality of the parameter space, i.e., the number of parameters being sampled.

3. **Compute acceptance ratio:**

$$r = \frac{P(\theta' | D)}{P(\theta^{(i)} | D)} = \frac{L(D | \theta') \pi(\theta')}{L(D | \theta^{(i)}) \pi(\theta^{(i)})} \quad (29)$$

where $L(D | \theta)$ is the likelihood and $\pi(\theta)$ is the prior.

4. **Accept or reject:** Draw u where u is random number in range $(0, 1)$. If $u < \min(1, r)$, accept the new state:

$$\theta^{(i+1)} = \theta', \quad \text{otherwise} \quad \theta^{(i+1)} = \theta^{(i)} \quad (30)$$

5. **Repeat** steps 2–4 for N iterations.

4.1.2 Posterior Estimation via Sampling

After discarding the initial *burn-in* samples, the remaining chain $\{\theta^{(i)}\}_{i=B+1}^N$ (with B being the burn-in cutoff) is used to estimate expectations under the posterior:

$$\mathcal{E}[f(\theta)] \approx \frac{1}{N-B} \sum_{i=B+1}^N f(\theta^{(i)}) \quad (31)$$

4.1.3 Acceptance Rate and Step Size

The step size of the proposal distribution q significantly affects the acceptance rate. If the steps are too small, the chain moves slowly; if too large, most proposals are rejected. For a model with d parameters, an optimal acceptance rate is typically in the range:

$$r_{\text{optimal}} \approx \frac{0.234}{\sqrt{d}} \quad \text{to} \quad 0.3 \quad (32)$$

For example, with $d = 4$ parameters, this suggests an optimal rate near 23%.

4.2 Pantheon+SH0ES Supernova Data

In this work, we use the Pantheon+ supernova dataset, a large and well-calibrated collection of Type Ia supernovae, which are used to study the expansion of the universe. The dataset provides the redshift and observed distance modulus for each supernova, corrected using light-curve data and host galaxy properties.

We use these observed distance moduli, μ_{obs} , along with their redshifts, to compare against the theoretical distance modulus $\mu_{\text{theory}}(z; \theta)$, which is computed using our cosmological model. This comparison forms the basis for our likelihood function in the MCMC analysis.

All data, including redshifts, distance moduli, and covariance matrices, are publicly available at: <https://github.com/PantheonPlusSH0ES/DataRelease>.

4.2.1 Distance Modulus in Standard Λ CDM

In the standard flat Λ CDM cosmological model, The Hubble parameter $H(z)$ as a function of redshift z is given by:

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}, \quad (33)$$

where H_0 is the Hubble constant today, Ω_m is the present matter density parameter, and Ω_Λ is the cosmological constant density parameter. In a spatially flat and $z < 5$ universe, we have $\Omega_m + \Omega_\Lambda = 1$.

The comoving distance to a redshift z is computed as:

$$\chi(z) = c \int_0^z \frac{dz'}{H(z')}, \quad (34)$$

where c is the speed of light.

For a flat universe, the luminosity distance $D_L(z)$ is related to the comoving distance by:

$$D_L(z) = (1 + z) \chi(z). \quad (35)$$

The theoretical distance modulus $\mu_{\text{theory}}(z)$, which relates the luminosity distance to the observed magnitude of standard candles such as Type Ia supernovae, is given by:

$$\mu_{\text{theory}}(z) = 5 \log_{10} \left(\frac{D_L(z)}{10 \text{ pc}} \right). \quad (36)$$

In practice, $D_L(z)$ is expressed in units of megaparsecs (Mpc), so Eq. (36) becomes:

$$\mu_{\text{theory}}(z) = 5 \log_{10} (D_L(z) [\text{Mpc}]) + 25. \quad (37)$$

This expression is used to compare theoretical predictions to observational data from supernovae surveys, such as Pantheon+, in order to constrain cosmological parameters through statistical methods like Markov Chain Monte Carlo (MCMC).

4.2.2 Distance Modulus in MTEGR

In order to obtain the theoretical distance modulus μ_{MTEGR} and compare it with the observed values μ_{Observed} , we make use of the redshift values corresponding to each supernova and solve equation 37. To proceed, we consider the first Friedmann equation in the framework of (MTEGR):

$$H^2 = \frac{8\pi G}{3(1 + S)} (\rho_M + \rho_{DE}) \quad (38)$$

From the previous detailed discussion, we know that for any type of matter, the density evolves as

$$\rho(z) = \rho_0 \frac{1 + S_0}{1 + S(z)} (1 + z)^{3(1+w)}. \quad (39)$$

By setting $z = 0$ in equation 38, we obtain

$$H_0^2 = \frac{8\pi G}{3(1 + S_0)} (\rho_{M0} + \rho_{DE0}).$$

If we assume that this matches the standard Λ CDM, given by

$$(H_0^{\Lambda\text{CDM}})^2 = \frac{8\pi G}{3} (\hat{\rho}_{M0} + \hat{\rho}_{DE0}),$$

we must then have

$$\frac{1}{(1 + S_0)} (\rho_{M0} + \rho_{DE0}) = (\hat{\rho}_{M0} + \hat{\rho}_{DE0}).$$

Therefore, we choose $\rho_0 = (1 + S_0) \hat{\rho}_0$ for all fluid components. Substituting this into the Friedmann equation yields:

$$\begin{aligned}
H^2 &= \frac{8\pi G}{3(1+S)}(\rho_M + \rho_{DE}) \\
&= \frac{8\pi G}{3(1+S)} \left(\rho_{M0} \frac{1+S_0}{1+S} (1+z)^3 + \rho_{DE0} \frac{1+S_0}{1+S} \right) \\
&= \frac{8\pi G(1+S_0)^2}{3(1+S)^2} (\hat{\rho}_{M0}(1+z)^3 + \hat{\rho}_{DE0}) \\
&= H_0^2 \left(\frac{1+S_0}{1+S} \right)^2 \left(\frac{\hat{\rho}_{M0}}{\hat{\rho}_{M0} + \hat{\rho}_{DE0}} (1+z)^3 + \frac{\hat{\rho}_{DE0}}{\hat{\rho}_{M0} + \hat{\rho}_{DE0}} \right) \\
&= H_0^2 \left(\frac{1+S_0}{1+S} \right)^2 (\Omega_{M0}(1+z)^3 + \Omega_{DE0}).
\end{aligned} \tag{40}$$

In the second line above, we assume $\omega_M = 0$ and $\omega_{DE} = -1$. The third line uses the previously defined rescaling $\rho_0 = (1+S_0)\hat{\rho}_0$, and in the fourth line, we introduce H_0 . Finally, in the last line, we define the density parameters $\Omega_i = \hat{\Omega}_i$, as both expressions are normalized to the same $(1+S_0)$.

We adopt the specific form $S(z) = \alpha(1+z)^\beta$, and to compute the distance modulus, we aim to solve equation 34 numerically:

$$\begin{aligned}
D_L &= c(1+z) \int_0^z \frac{dz'}{H(z')} \\
&= \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{\left(\frac{1+S_0}{1+\alpha(1+z')^\beta} \right) \sqrt{\Omega_{M0}(1+z')^3 + 1 - \Omega_{M0}}}.
\end{aligned} \tag{41}$$

We note that the luminosity distance $D_L = D_L(z, \alpha, \beta, \Omega_{M0}, H_0)$, which implies that we can solve the integral numerically for any given set of parameter values. Consequently, the parameter samples used in equation 27 are given by $\theta = [\alpha, \beta, \Omega_{M0}, H_0]$.

4.2.3 Tuning the Proposal Distribution

In implementing the Markov Chain Monte Carlo (MCMC) algorithm, it is essential to appropriately tune the proposal distribution to ensure efficient sampling. In our code, we perform tuning by adjusting the scale of the standard deviation, σ_i in 28.

The scale σ controls the size of the steps taken in the parameter space during the MCMC process. A very small scale may lead to slow exploration of the parameter space (high acceptance rate but poor mixing), while a very large scale can cause a high rejection rate (poor acceptance rate).

To find an optimal configuration, we experimented with different values for the scale of sigma and monitored the acceptance rate. Our goal was to achieve an acceptance rate close to 25%, which is typically considered efficient for high-dimensional problems with uncorrelated proposals.

4.3 MCMC for SNe

After tuning the MCMC sampler and running the chains with the final configuration, we analyze the posterior distributions of the model parameters. The goal is to estimate the best-fit

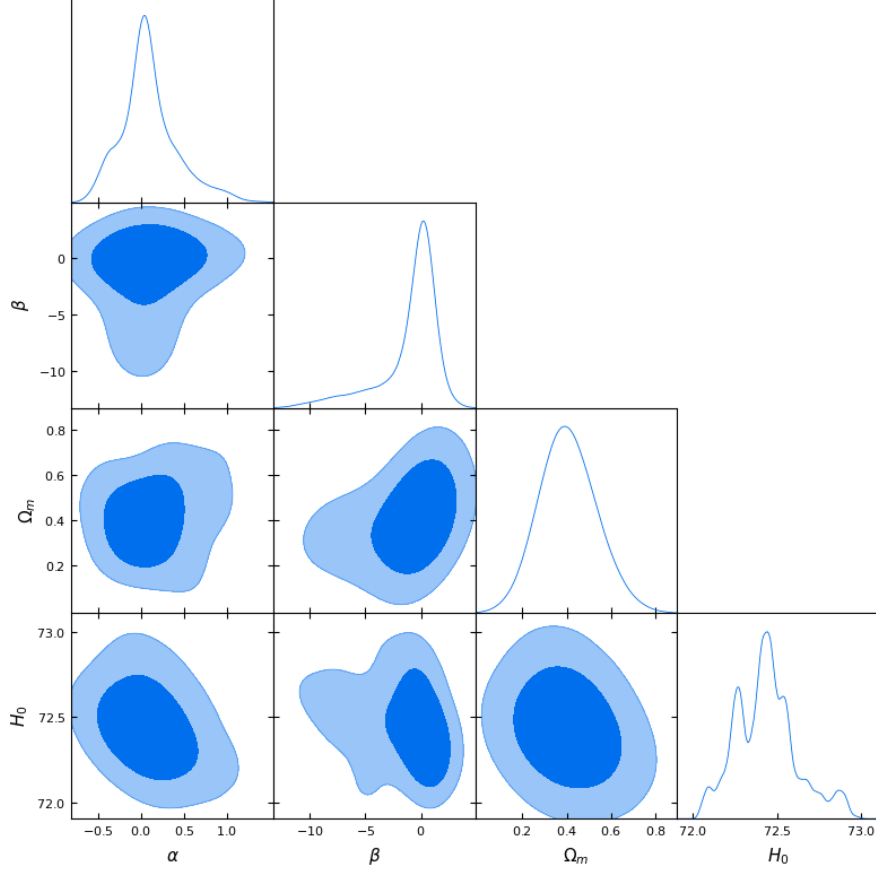


Figure 2: Triangle plot showing the 1D and 2D marginal posterior distributions of the model parameters.

values along with their uncertainties and the correlations between parameters.

Table 2 and figure ?? summarizes the results of the MCMC analysis which inferred from the posterior distributions.

Table 1: Posterior estimates for the MTEGR model parameters.

Parameter	Mean	Standard Deviation
α	0.52	0.46
β	-0.99	2.46
Ω_{M0}	0.3392	0.1528
H_0	72.50	0.31

4.4 MCMC for DESI DR2

We implemented a MCMC sampler in `Python` to constrain the parameters of MTEGR model using Baryon Acoustic Oscillation data. The likelihood function was based on the χ^2 between the observed and theoretical volume-averaged distances $D_V(z)/r_d$. After tuning, the final MCMC chain contained 1,00,000 samples with a total acceptance rate of **23%**.

Table 2: Posterior estimates for the MTEGR model parameters.

Parameter	Mean	Standard Deviation
α	0.470	0.961
β	-2.433	2.941
Ω_M	0.267	0.093
H_0 [km/s/Mpc]	67.49	2.490

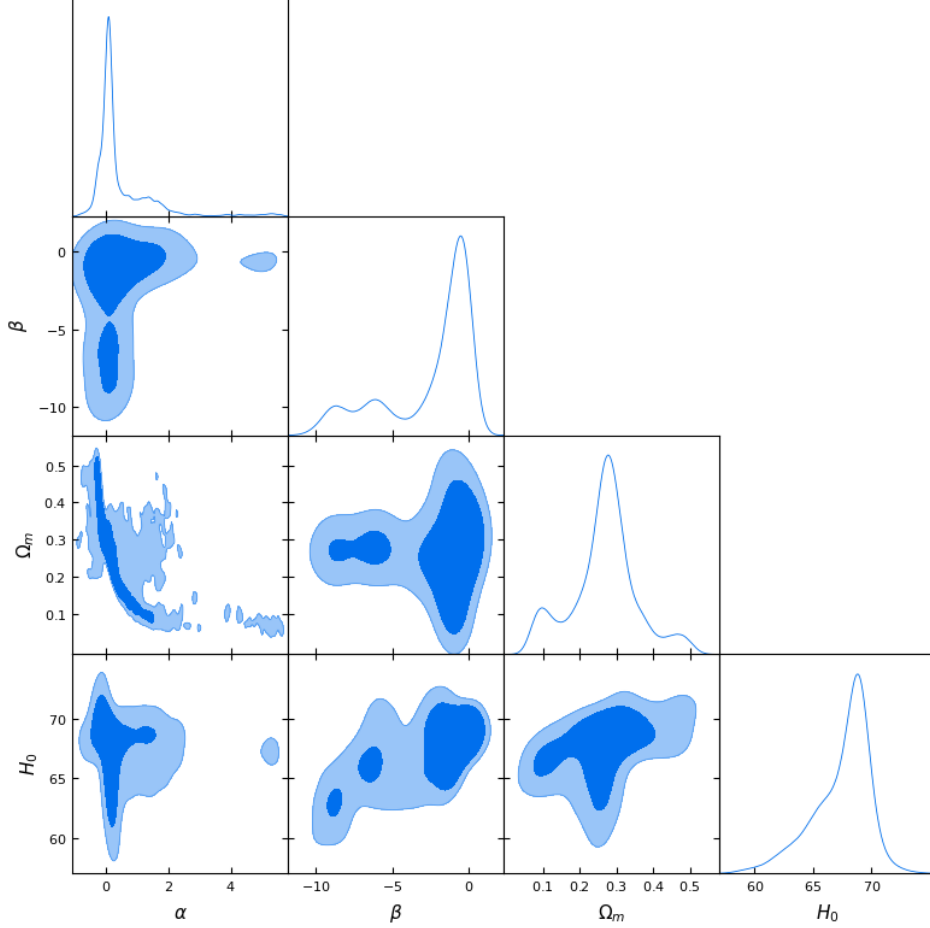


Figure 3: Triangle plot of BAO samples.

4.5 MCMC Analysis for BAO+SNe

To improve the constraints on cosmological parameters, we combine the Supernovae and Baryon Acoustic Oscillations data. This is done by constructing a total chi-squared function as the sum of the individual chi-squared contributions from each dataset:

$$\chi_{\text{total}}^2 = \chi_{\text{SNe}}^2/1700 + \chi_{\text{BAO}}^2/9. \quad (42)$$

This approach assumes equal weight (or importance) for both datasets and allows us to jointly constrain the parameters using a single MCMC analysis.

In our analysis, the MCMC chains for the combined dataset did not converge. This could be attributed to several factors: a relatively short runtime (limited number of MCMC steps), suboptimal choice of initial parameters or step sizes, or a possible incompatibility of the model with both datasets simultaneously.

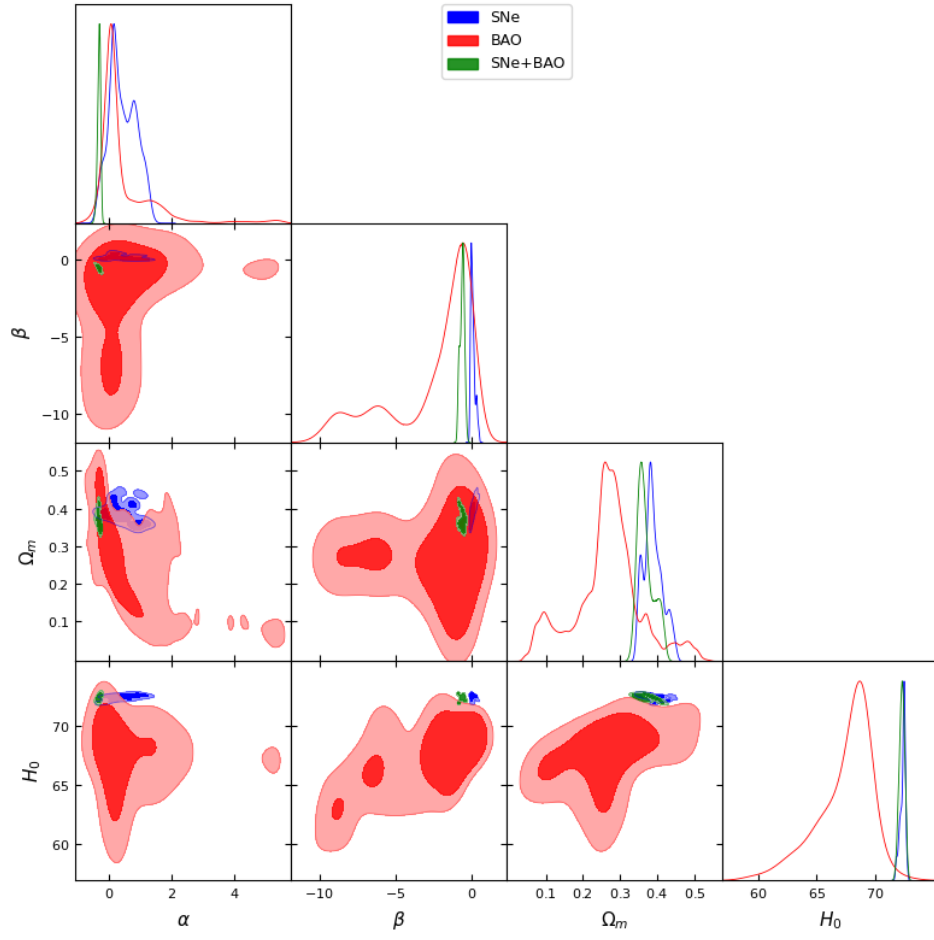


Figure 4: Triangle plot of BAO samples.